



TRINITY GRAMMAR SCHOOL

Mathematics Department

2012

TASK 3 - HALF YEARLY
EXAMINATION

Year 12

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Weighting 30%
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 11 – 16.
- Write your Board of Studies Student Number **and** Class Teacher on the writing booklet(s) **or** sheet(s) submitted.

Total marks – 100

Section I

Pages 2 – 4

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II

Pages 5 – 11

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

SECTION 1 Total marks (10)

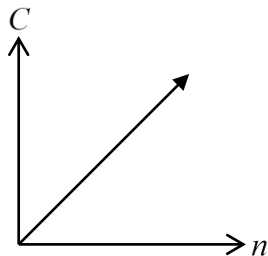
- Circle the correct response on the answer sheet provided.
 - Each question is worth 1 mark.
-

- 1 A ship sails 9 nautical miles north, then 12 nautical miles east. What is the ship's bearing from its starting point?

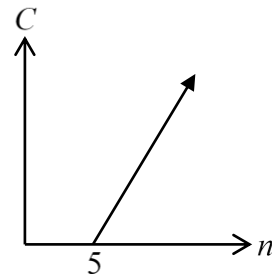
(A) 037° (B) 053° (C) 217° (D) 233°

- 2 The cost, C , of producing textbooks is given by the formula $C = 14n + 5$. Which of the graphs below best represents this equation?

(A)

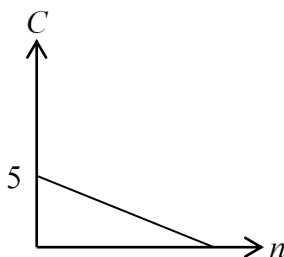


(B)

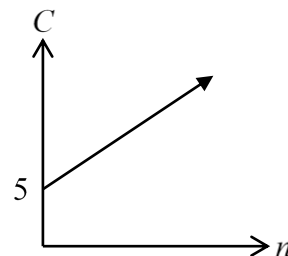


Not to scale

(C)



(D)

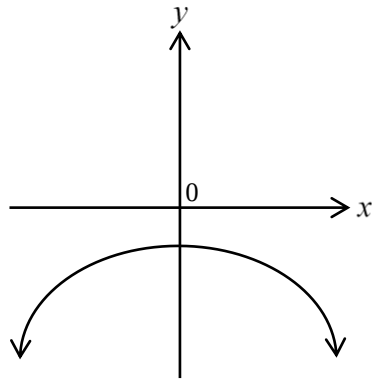


- 3 The force, F , on a particular moving object, moving with a velocity of V , is defined by the equation $F = \frac{3V^2}{4}$

Which of the following could be a value for V if the value of F is 12?

(A) 3 (B) 4 (C) 9 (D) 16

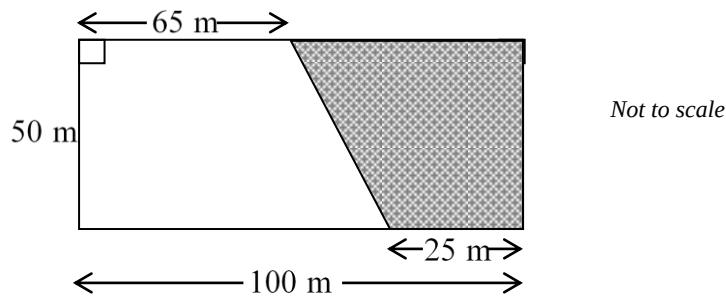
4



The parabola shown could have the equation.

- (A) $y = x^2 - 3$ (B) $y = -x^2 + 3$ (C) $y = -x^2 - 3$ (D) $y = x^2 + 3$

5



The diagram shows a yard that consists of a rectangle split into two trapezia.

Find the shaded area to the nearest square metre.

- (A) 1500 (B) 2500 (C) 3500 (D) 5000

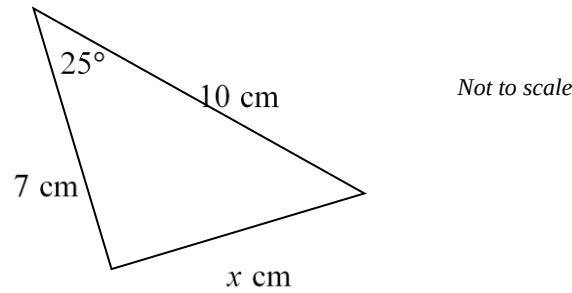
6 Find $\log_{10} 4^3$ correct to 3 significant figures.

- (A) 0.218 (B) 1.806 (C) 0.22 (D) 1.81

7 $16 - 4(x - 5) =$

- (A) $12x + 60$ (B) $12x - 60$ (C) $-4 - 4x$ (D) $36 - 4x$

8



Which of the following equations should be used to find the value of x in the triangle above?

(A) $\tan 25^\circ = \frac{x}{10}$

(B) $x^2 + 7^2 = 10^2$

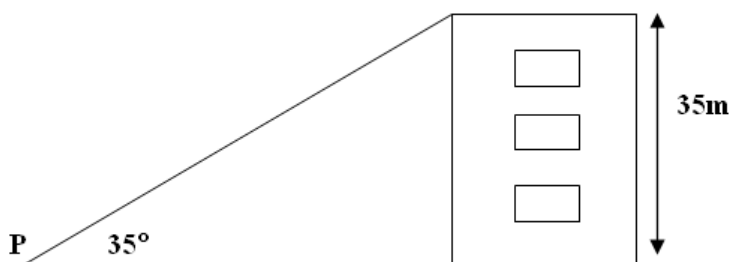
(C) $x^2 = 7^2 + 10^2 - 2 \times 7 \times 10 \times \cos 25^\circ$

(D) $\frac{x}{\sin 25} = \frac{7}{\sin 10^\circ}$

- 9 The water from a flat rectangular roof 12 metres by 15 metres is collected in a cylindrical tank of diameter 2.2 metres. What rise in water level in the tank is caused by rainfall of 1 cm ?

- (A) 0.12 m (B) 0.47 m (C) 12 m (D) 47 m

10



The angle of elevation of the top of a building from a point P on the ground is 35° . If the building is 35 metres tall, find the distance from the base of the building to the point P, correct to the nearest metre.

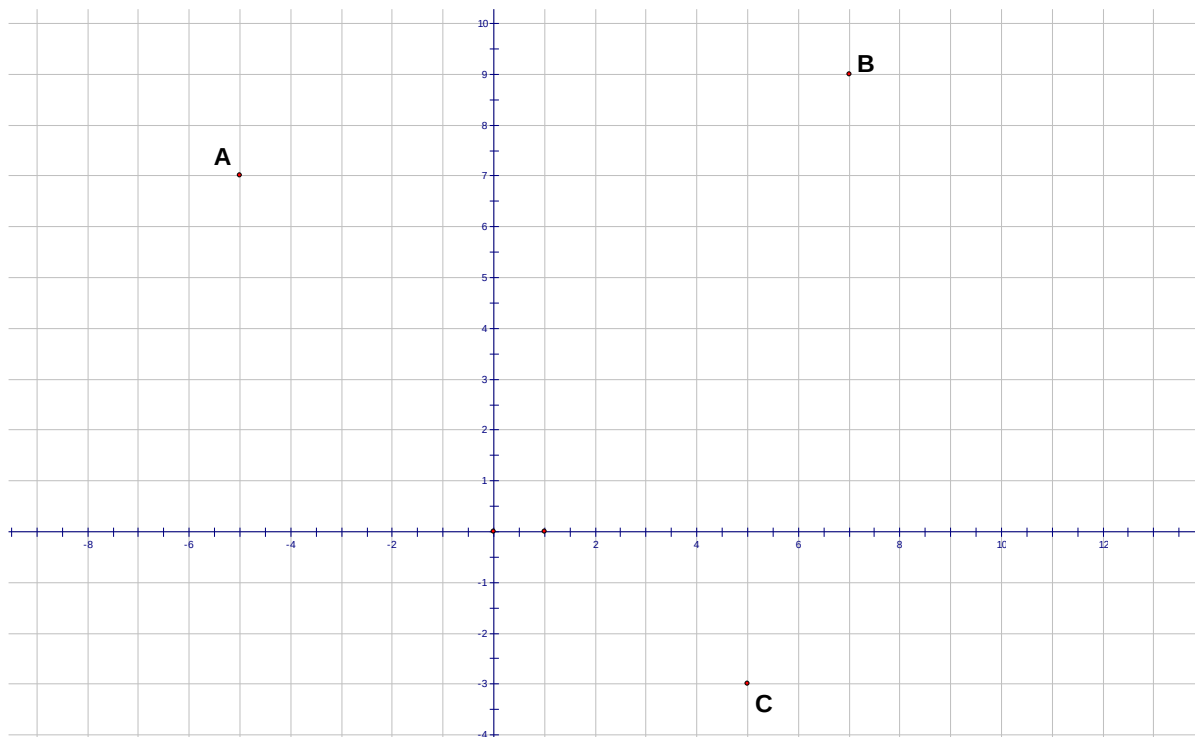
- (A) 20 (B) 25 (C) 43 (D) 50

SECTION 2 Total marks (90)

- Begin each question in a new writing booklet or on a new answer sheet.
- Show all necessary working.
- Each question is worth 15 marks.

Question 11 (15 marks)	Use a SEPARATE writing booklet	Marks
(a)	Evaluate $2e^4$, correct to three significant figures.	2
(b)	Solve $2x^2 = x$	2
(c)	Factorise fully $8x^3 - 1000$	2
(d)	Simplify $\frac{6x^2 + 5x - 6}{4x^2 + 4x - 3}$	2
(e)	Evaluate $ -3 \times 4 - 2 \times -5 $	2
(f)	Find the value of $\log_2 7$ correct to 3 decimal places.	2
(g)	Find the points of intersection of the functions; $x^2 - 4x + y = 0 \text{ and } y = -2x + 5$	3

- (a) The points A(-5, 7), B(7, 9) and C(5,-3) are shown on the diagram below.



- | | |
|---|---|
| (i) Show that AB has equation $x - 6y + 47 = 0$. | 2 |
| (ii) Show that the length of AB = the length of BC. | 2 |
| (iii) Show that the midpoint, M, of AC is (0, 2). | 1 |
| (iv) Find l , the equation of the perpendicular bisector of AC. | 2 |
| (v) Show that l passes through B. | 1 |
| (vi) Find the perpendicular distance from M to AB. | 2 |
| (vii) Find the area of $\triangle ABC$. | 2 |
- (b) Find the equation of the normal to the curve $y = (2e^x + 1)^2$ at the point where $x = 0$

3

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Convert $\frac{3\pi}{2}$ radians into degrees.

1

(b) Differentiate each of the following with respect to x .

(i) $y = 3e^{4x-1} + \frac{1}{e^x}$

2

(ii) $y = 2x \ln x$

2

(iii) $y = \ln(e^x + 1)$

2

(iv) $y = e^x + \ln x^3$

2

(c) Evaluate $\int_0^2 3x - 1^3 dx$

3

(d) (i) Solve $x^2 - 9x \geq 0$

2

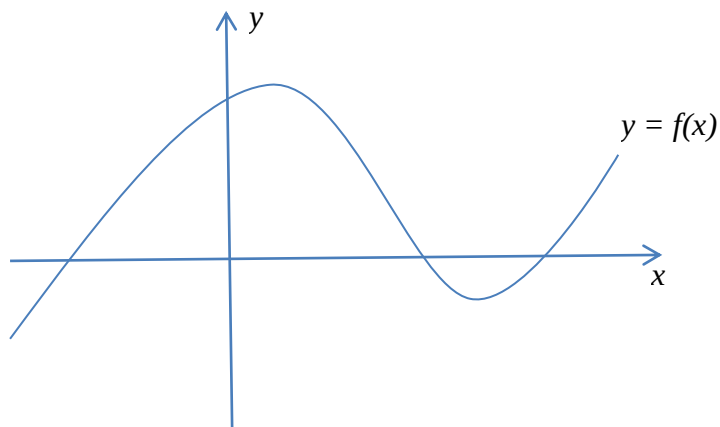
(ii) Hence or otherwise find the domain of $y = \ln(x^2 - 9x)$

1

Question 14 (15 marks) Use a SEPARATE writing booklet.
Marks

(a)

2



Copy or trace the function above onto your page. On the same axes, sketch a possible graph of the gradient function $f'(x)$

(b)

(i) Copy and complete the table below correct to 2 decimal places

2

x	1	1.5	2	2.5	3
$y = 1 + 2 \ln x$					

(ii) Use the five function values from part i) and 2 applications of Simpson's Rule to estimate the value of

2

$$\int_1^3 1 + 2 \ln x \, dx$$

(c) Find

(i) $\int \left(\frac{1}{x} + x \right) dx$

2

(ii) $\int \sqrt{2x+5} \, dx$

1

(iii) $\int \frac{3x^2 + 2x + 1}{x^2} dx$

2

(d) Given that $\log_a 5 = x$ and $\log_a 3 = y$

find an expression in terms of x and y for

(i) $\log_a 75$ **1**

(ii) $\log_a 0.6$ **1**

(e) Find the exact arc length of a circle of radius is 4cm and angle subtended is 120° . **2**

Question 15 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The equation $3x^2 - 8x + 9 = 0$ has roots α and β

Evaluate:

(i) $\alpha + \beta$ **1**

(ii) $\alpha \beta$ **1**

(iii) $\alpha^2 + \beta^2$ **1**

- (b) Consider the function $y = 4x^3 + 6x^2 - 144x + 1$

(i) Find the stationary points and determine their nature **3**

(ii) Find the point of inflexion **1**

(iii) Find the maximum value of the function in the domain $-5 \leq x \leq 7$ **1**

(iv) Sketch the graph of $y = 4x^3 + 6x^2 - 144x + 1$ in this domain **2**

- (c) A piece of wire 24 cm long is cut into 2 pieces. One of the pieces is bent into a square with sides of x cm. The other piece is bent to form a right angled isosceles triangle. The equal sides of the triangle are of length y cm.

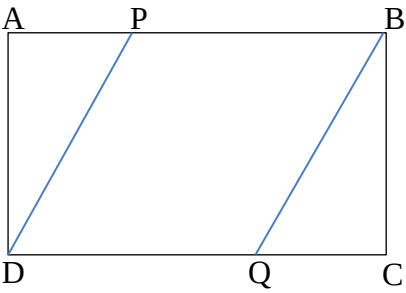
(i) Show that $y = \frac{24 - 4x}{2 + \sqrt{2}}$ **2**

(ii) Find an exact expression for the total area covered by the square and the triangle. **1**

(iii) Find the exact value for x so that the total area is a minimum. **2**

Question 16 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) For the parabola with equation $x^2 - 6x - 8y + 17 = 0$ find,
- (i) the co-ordinates of the vertex. 2
 - (ii) the co-ordinates of the focus. 1
 - (iii) the equation of the axis of symmetry. 1
- (b) The arc of the curve $y = e^x - 1$ between $x = 1$ and $x = 3$ is rotated about the x axis. Calculate the volume of the solid generated giving your answer to 3 significant figures. 3
- (c) 
- In the diagram, ABCD is a rectangle. The points P and Q are on the intervals AB and CD respectively, and $AP = CQ$.
- (i) Prove that the triangles APD and BCQ are congruent. 2
 - (ii) Prove that PBQD is a parallelogram. 2
- (d) The fifth term of an arithmetic series is 14 and the sum of the first 10 terms is 165. Find the first term of the series. 3
- (e) Find the limiting sum of $6 + 3 + 1\frac{1}{2} + \dots$ 1

-- End of Paper --

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$



TRINITY GRAMMAR SCHOOL
2012, Year 12 Mathematics
Half Yearly Examination
SECTION I
ANSWER SHEET

Name:

Class Teacher:

Be sure to write your answers for Section I on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. Mark only one answer for each question.

- | | | | | |
|------|---|---|---|---|
| Q1. | A | B | C | D |
| Q2. | A | B | C | D |
| Q3. | A | B | C | D |
| Q4. | A | B | C | D |
| Q5. | A | B | C | D |
| Q6. | A | B | C | D |
| Q7. | A | B | C | D |
| Q8. | A | B | C | D |
| Q9. | A | B | C | D |
| Q10. | A | B | C | D |



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- | | | | | |
|------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Q1. | A | ☒ B | C | D |
| Q2. | A | B | C | ☒ D |
| Q3. | A | ☒ B | C | D |
| Q4. | A | B | ☒ C | D |
| Q5. | ☒ A | B | C | D |
| Q6. | A | B | C | ☒ D |
| Q7. | A | B | C | ☒ D |
| Q8. | A | B | ☒ C | D |
| Q9. | A | ☒ B | C | D |
| Q10. | A | B | C | ☒ D |

Question 11.

a) $109. \checkmark \sqrt{(3 \text{ s.f})} \quad 2$

b) $2x^2 - x = 0$

$$x(2x - 1) = 0 \checkmark$$

$$x = 0/x = \frac{1}{2} \checkmark \quad 2$$

c) $8x^3 - 1000 = 8(x^3 - 125) \quad \text{OR} \quad (2x-10)(4x^2+20x+100)$
 $= 8(x^3 - 5^3)$
 $= 8(x-5)(x^2+5x+25) \quad 2$

d) $\frac{6x^2 + 5x - 6}{4x^2 + 4x - 3} = \frac{6x^2 + 9x - 4x - 6}{4x^2 + 6x - 2x - 3}$
 $= \frac{3x(2x+3) - 2(2x+3)}{2x(2x+3) - 1(2x+3)}$
 $= \frac{(3x-2)(2x+3)}{(2x-1)(2x+3)} \checkmark$
 $= \frac{3x-2}{2x-1} \checkmark \quad 2$

e) $|-12| - |-10| = 12 - 10 = 2 \checkmark \quad 2$

f) $\log_2 7 = \frac{\log_{10} 7 \checkmark \leftarrow 0.845}{\log_{10} 2 \leftarrow 0.301} \checkmark = 2.807 \checkmark \quad 2$

g) $x^2 - 4x + y = 0$
 $x^2 - 4x - 2x + 5 = 0$

$$x^2 - 4x - 2x + 5 = 0$$

$$x^2 - 6x + 5 = 0$$

$$(x-5)(x-1) \checkmark = 0$$

$$x = 5; x = 1 \checkmark$$

When $x=1$, $y = -2+5=3$

When $x=5$, $y = -10+5=-5 \checkmark$

$\therefore$ Pts of intersection
are $(1, 3)$ and $(5, -5)$
 $\underline{\underline{2}}$

Question 12

$$a) (i) m_{AB} = \frac{9-7}{7-5} = \frac{2}{12} = \frac{1}{6} \quad (1)$$

$$y-7 = \frac{1}{6}(x+5)$$

$$6y-42 = x+5 \quad (1)$$

$$x-6y+47=0.$$

2

$$(ii) d_{AB} = \sqrt{(7-5)^2 + (9-7)^2}$$
$$= \sqrt{12^2 + 2^2}$$
$$= \sqrt{148} \quad (1)$$

$$d_{BC} = \sqrt{(5-7)^2 + (-3-9)^2}$$
$$= \sqrt{(-2)^2 + (-12)^2}$$
$$= \sqrt{148} \quad (1)$$

$$\therefore d_{AB} = d_{BC}$$

2

$$(iii) M_{AC} = \left(\frac{-5+5}{2}, \frac{7+(-3)}{2} \right) \quad (1)$$
$$= (0, 2)$$

1

$$(iv) m_{AC} = \frac{-3-7}{5-5} = \frac{-10}{10} = -1$$

$$\therefore \text{Gradient of normal} = 1 \quad (1)$$

$$y-2 = 1(x-0)$$

$$y = x+2$$

$$x-y+2=0$$

(1)

2

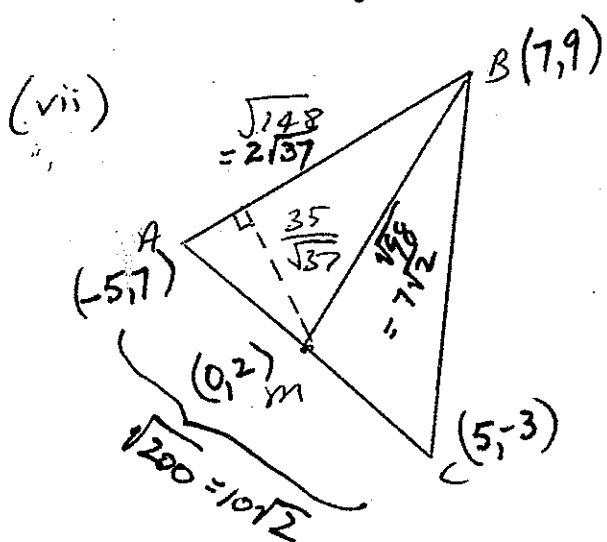
(v) If $y = x + 2$ and $x = 7$, $y = 7 + 2 = 9$
 $\therefore B(7, 9)$ is on l . (1)

(vi)
$$d = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|1 \times 0 - 6 \times 2 + 47|}{\sqrt{1^2 + 6^2}} \quad (M1)$$

$$= \frac{35}{\sqrt{37}} \quad (1)$$

2



$$A = \frac{1}{2} \times \frac{35}{\sqrt{37}} \times \frac{2\sqrt{37}}{\sqrt{148}} \times 2 \quad (M1)$$

$$= 70 \text{ sq units} \quad (1)$$

2

b)
$$y = (2e^x + 1)^2$$

$$y' = 2(2e^x + 1) \times 2e^x$$

$$= 4e^x(2e^x + 1) \quad (1)$$

$$= 12 \text{ when } x = 0$$

$\therefore$ Gradient of normal $= -\frac{1}{12}$

$$y - 9 = -\frac{1}{12}(x - 0) \quad (1)$$

$$12y - 108 = -x$$

$$x + 12y - 108 = 0 \quad (1)$$

$$\text{OR } y = -\frac{1}{12}x + 9$$

3

Question 13

a) $\frac{3\pi}{2} \times \frac{180}{\pi} = 270^\circ$

1

b) (i) $y = 3e^{4x-1} + \frac{1}{e^x}$

$$y' = 12e^{4x-1} - \frac{1}{e^x}$$

2

(ii) $y = 2x \ln x$

$$y' = \ln x \times 2 + 2x \times \frac{1}{x}$$

$$= 2 \ln x + 2$$

2

(iii) $y = \ln(e^x + 1)$

$$y' = \frac{e^x}{e^x + 1}$$

2

(iv) $y = (e^x + \ln x)^3$

$$y' = 3(e^x + \ln x)^2 \times (e^x + \frac{1}{x})$$

$$= 3(e^x + \frac{1}{x})(e^x + \ln x)^2$$

2

c) $\int_0^2 (3x-1)^3 dx = \frac{1}{12} (3x-1)^4 \Big|_0^2$

$$= \frac{1}{12} \times 5^4 - \frac{1}{12} (-1)^4$$

$$= 52$$

3

d) (i) $x^2 - 9x \geq 0$

$$x(x-9) \geq 0$$

$$x \leq 0, x \geq 9$$

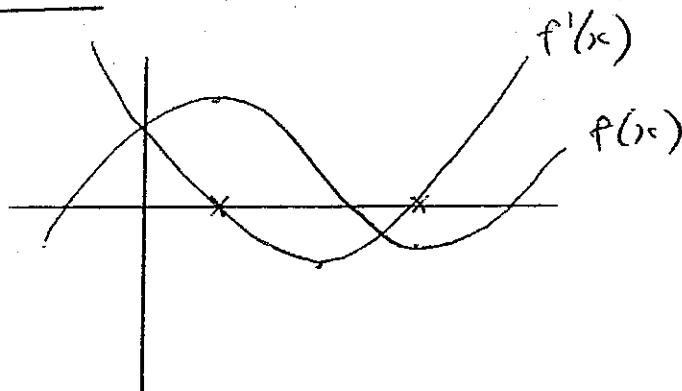
2

(ii) The domain is $x < 0, x > 9$

1

Question 14

a)



2

b)

x	1	1.5	2	2.5	3
y = 1 + 2Ln x	1	1.81	2.39	2.83	3.20

(i)

2

(ii) $\int_1^3 1 + 2 \ln x \, dx \doteq \frac{0.5}{3} [(1+3.2) + 4(1.81+2.83) + 2(2.39)]$
 $= 4.59$ (1 decimal place) 2

$$c) \quad (1) \quad \int \frac{1}{x} + x \, dx = \ln x + \frac{1}{2} x^2 + C \quad 2$$

$$\begin{aligned} \text{(ii)} \quad \int (2x+5)^{\frac{1}{2}} dx &= \frac{2}{3} \left(\frac{2x+5}{2} \right)^{\frac{3}{2}} + C \\ &= \frac{\sqrt{(2x+5)^3}}{3} + C \end{aligned}$$

$$(ii) \int 3 + 2x^{-1} + x^{-2} dx = 3x + 2 \ln x - \frac{1}{x} + C \quad \text{① progress} \quad \text{①}$$

d) (i) $\log_a 75 = \log_a 5^2 + \log_a 3$
 $= 2 \log_a 5 + \log_a 3$
 $= 2x + y$

$$\begin{aligned} \text{(ii)} \quad \log_2 0.6 &= \log_2 \frac{3}{5} \\ &= \log_2 3 - \log_2 5 \\ &= y - x \end{aligned}$$

e)

$$l = r\theta$$

$$= 4 \times 120 \times \frac{\pi}{180}$$

$$= 4 \times \frac{2\pi}{3}$$

$$= \frac{8\pi}{3} \text{ cm}$$

2

if not exact form?

Question 15

$$a) (i) \alpha + \beta = \frac{-b}{a} = -\frac{(-8)}{3} = \frac{8}{3} \quad \checkmark$$

$$(ii) \alpha\beta = \frac{c}{a} = \frac{9}{3} = 3 \quad \checkmark$$

$$\begin{aligned} (iii) \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= \left(\frac{8}{3}\right)^2 - 2 \times 3 \\ &= 1\frac{1}{9} \quad \checkmark \end{aligned}$$

$$b) y = 4x^3 + 6x^2 - 144x + 1$$

$$(i) \frac{dy}{dx} = 12x^2 + 12x - 144 = 0 \quad \text{for stat. pts}$$

$$12(x^2 + x - 12) = 0$$

$$12(x+4)(x-3) = 0$$

$$x = -4, 3 \quad \checkmark$$

$$\text{If } x = -4, y = 417$$

$$\text{If } x = 3, y = -269$$

$$\frac{d^2y}{dx^2} = 24x + 12 < 0 \quad \text{when } x = -4$$

$$\therefore (-4, 417) \text{ is a max } \checkmark$$

$$\frac{d^2y}{dx^2} = 24x + 12 > 0 \quad \text{when } x = 3$$

$$\therefore (3, -269) \text{ is a min } \checkmark$$

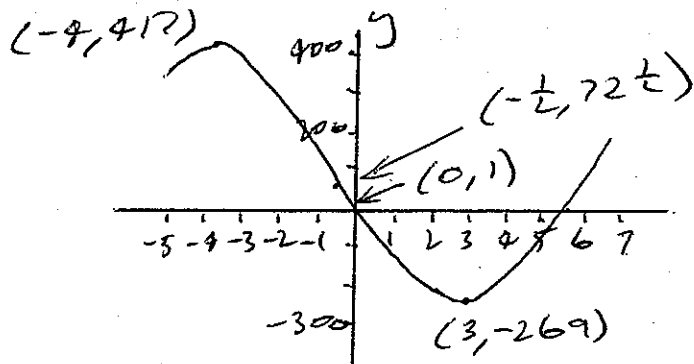
$$(ii) \text{ Pt of inflexion exists when } \frac{d^2y}{dx^2} = 0$$

$$\begin{aligned} \therefore 24x + 12 &= 0 \\ x &= -\frac{1}{2} \quad \checkmark \end{aligned}$$

$$\text{Pt of inflexion at } \left(-\frac{1}{2}, 74\right) \quad \checkmark$$

$$(iii) \text{ Max when } x = 7, y = 4(7)^3 + 6(7)^2 - 144(7) + 1 = 659$$

(iv)



c) (i) $4x + 2y + \sqrt{2}y = 24$
 $4x + (2 + \sqrt{2})y = 24$
 $y = \frac{24 - 4x}{2 + \sqrt{2}}$

(ii) $A = x^2 + \frac{1}{2}y^2$
 $= x^2 + \frac{1}{2} \left(\frac{24 - 4x}{2 + \sqrt{2}} \right)^2$

(iii) $\frac{dA}{dx} = 2x + 2x \cdot \frac{1}{2} \times \left(\frac{-4}{2 + 2\sqrt{2}} \right) \left(\frac{24 - 4x}{2 + 2\sqrt{2}} \right) = 0$

$$2x - \frac{4(24 - 4x)}{(2 + 2\sqrt{2})^2} = 0$$

$$2x(2 + 2\sqrt{2})^2 = 4(24 - 4x)$$

$$2x(2 + 2\sqrt{2})^2 = 96 - 16x$$

$$16x + 2x(2 + 2\sqrt{2})^2 = 96$$

$$x = \frac{96}{16 + 2(2 + 2\sqrt{2})^2}$$

$$= 3.74 \text{ (correct to 2 dp)}$$

Question 16

a) $x^2 - 6x - 8y + 17 = 0$

$$(x-3)^2 - 9 = 8y - 17$$

$$(x-3)^2 = 8y - 8$$

$$(x-3)^2 = 8(y-1)$$

(i) Vertex at $(3, 1)$

2

(ii) Since $4a = 8$
 $\therefore a = 2$

$\therefore$ Focus at $(3, 3)$

1

(iii) Axis of symmetry at $x = 3$

1

b) $V = \pi \int_1^3 (e^x - 1)^2 dx$ ✓

$$= \pi \int_1^3 (e^{2x} - 2e^x + 1) dx$$

$$= \pi \left[\frac{1}{2} e^{2x} - 2e^x + x \right]_1^3 \checkmark$$

$$= \pi \left[\left(\frac{1}{2} e^6 - 2e^3 + 3 \right) - \left(\frac{1}{2} e^2 - 2e + 1 \right) \right]$$

$$= 519 \text{ (correct to 3 sig. fig)} \checkmark$$

3

c) (i) $\triangle APD \equiv \triangle BQC$ (SAS test)

2

(ii) $DP = BQ$ (corresponding sides of congruent triangles)

$PB = DQ$ Since $AB = DC$ (opp sides of rect.)
and $AP = QC$ (Given)

$\therefore$ $PBQD$ is a parallelogram since opposite sides are equal.

2

$$d) \quad T_5 = a + (5-1)d = 14$$

$$\therefore a + 4d = 14 \quad \checkmark$$

$$\text{Also, } S_{10} = \frac{10}{2} [2a + (10-1)d] = 165 \quad \checkmark$$

$$5(2a + 9d) = 165$$

$$2a + 9d = 33$$

Now solve simultaneously

$$a + 4d = 14 \quad (1)$$

$$2a + 9d = 33 \quad (2)$$

$$(1) \times 2$$

$$2a + 8d = 28$$

$$2a + 9d = 33$$

$$\therefore d = 5$$

$$\text{Sub } d=5, \text{ in } a + 4d = 14$$

$$a + 20 = 14$$

$$\therefore a = -6 \quad \checkmark \quad 3$$

$$e) \quad S_{\infty} = \frac{a}{1-r}$$

$$= \frac{6}{1-\frac{1}{2}}$$

$$= 12$$

1

